Prob. 7. (2&T. Ch. 7., 8.3)

Show that the const. term in the FS of $f(x)$ is equal to the average of $f(x)$ over the interval $[-\pi,\pi]$, given by:

(1) $$\frac{a_0}{2} = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

- Note:

This is very useful for checking your calculation of a_0: If you can visualize $f(x)$, you can look at your value of $a_0/2$ & see if it makes sense. So for example, if your f-n is always positive over $[-\pi,\pi]$, a_0 should be positive as well.

Soln: Recall the FS of $f(x)$:

(2) $$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\cos(nx) + b_n\sin(nx)\right]$$

Now let me multiply BS of (2) by 1, & integrate (S) it on $[-\pi,\pi]$:

(3) $$\int_{-\pi}^{\pi} f(x)\cdot 1\,dx = \int_{-\pi}^{\pi}\frac{a_0}{2}\cdot 1\,dx + \underbrace{\int_{-\pi}^{\pi}\sum_{n=1}^{\infty}\left[a_n\cos(nx) + b_n\sin(nx)\right]\cdot 1\,dx}$$

Let's assume "nice" cases in Pr. 5, (7)-(11) for $\sum_{n=1}^{\infty}$ to be uniform convergent. Then

(4) $$\text{RHS} = \int_{-\pi}^{\pi}\frac{a_0}{2}\,dx + \int_{-\pi}^{\pi}\sum_{n=1}^{\infty}\left[a_n\cos(nx) + b_n\sin(nx)\right]dx =$$

$$= \frac{a_0}{2}\cdot\left[x\right]_{-\pi}^{x=\pi} + \sum_{n=1}^{\infty}\left\{a_n\underbrace{\int_{-\pi}^{\pi}\cos(nx)\,dx}_{=0 \text{ by Pr. 5(a)}} + b_n\underbrace{\int_{-\pi}^{\pi}\sin(nx)\,dx}_{=0 \text{ by Pr. 5, (b)}}\right\} =$$

$$= \frac{a_0}{2}\left[2\pi\right] + 0 = a_0\cdot\pi.$$ $\Rightarrow$ Putting this RHS to (3):

$$\int_{-\pi}^{\pi} f(x)\,dx = a_0\cdot\pi \Rightarrow \boxed{a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx}$$ establishing the desired result.